

Programming Assignment #3

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Question #1

Our plaintext matrix needs to have columns with 4 elements so that it can be properly multiplied with our selected 4x4 invertible matrix. Technically, the matrix could have had more than 4 rows, as long as we selected a larger invertible matrix that matched the size.

Pseudocode

Import numpy

Part 1

```
File_path = Input("Enter file path")
```

```
file_contents = open(file_path)
```

```
matrix = []
```

```
while file_contents != "":
```

```
Matrix.append(ord(pop(file_contents[0])))
```

```
matrix *= matrix_a
```

```
print(matrix)
```

```
Print(matrix_a)
```

Part 2

```
File_path = Input("Enter file path")
```

```
file_contents = open(file_path)
```

```
matrix = []
```

```
while file_contents != "":
```

```
Matrix.append(pop(file_contents[0]))
```

```
matrix *= matrix_a_inverse
Decrypted_message = ""
For i in matrix.shape:
    For j in matrix_shape:
        Decrypted_message += chr(matrix[i, j])
Print(decrypted_message)
```